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Comparison of head-up display (HUD) vs. head-down display (HDD): driving performance of commercial vehicle operators in Taiwan

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Abstract

This study investigates the effects of two different display modes—head-up display (HUD) vs. head-down display (HDD) on the driving performance and psychological workload ratings of drivers operating commercial vehicles in Taiwan. Twelve commercial lorry drivers participated in a 2 (high/low driving load road) $\times 2$ (head-up/head-down display) $\times 2$ (different arrangements of display sequences used) mixed-factor driving simulation experiment. Participants were divided into two groups according to the level of driving load conditions within each driving load group; the participants were further divided into another 2 subgroups based on two arrangements of display sequences used. For each driving load condition, there were two 20-min driving simulation experiments, separated by a display sequence using head-up first and then head-down or vice versa. The subjects were asked to perform four tasks: “commercial goods delivery”, “navigation”, “speed detection and maintenance” and “response to an urgent event”. Results indicated that for the first task, commercial goods delivery, the two display types showed no significant performance difference in terms of average accuracy rate. However, in terms of response time to an urgent event, it was faster with the HUD (with a low driving load—head-up vs. head-down: 1.0073 vs. 1.8684 s; with a high driving load—head-up vs. head-down: 1.3235 vs. 2.3274 s) and speed control was more consistent (having low speed variations) than with the HDD. In addition, using the HUD caused less mental stress for the drivers than the HDD and was

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easier for first-time users to become familiar with; with a high driving load, however, the difference between the two displays was not significant.

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1. Introduction

Transport system logistics is a major factor in controlling time and cost in the transportation industry; in other words, the transportation of goods between two points in a quick, effective and safe manner is a crucial management issue in this industry. Against a background of rapid development in information technology (IT) and communications technology, numerous tools are available to assist in the design of a traditional transportation system. One such sub-system of Intelligent Transportation Systems (ITS) is the Commercial Vehicle Operation System (CVOS), the purpose of which is to upgrade the efficiency and safety of transportation through logistics (Collins et al., 1999).

To achieve these goals, CVO systems must provide drivers with large amounts of information from many categories (e.g. route guidance/navigation, traffic signs, cargo/road/vehicle conditions), and select the best way to display this information; important considerations include having a user-friendly system, since a driver's capacity to process this information is a key factor in its acceptance and use.

In recent years, many car manufacturers in Taiwan have introduced car information systems, such as the Ford e-car[®], and navigational systems, such as the Nissan Toobe[®]. Coincidentally, driving support systems, such as these, all use a 6–8 inch LCD as the information display interface, which is positioned in the middle of the vehicle's control panel (usually above or under the air conditioner and stereo controls). Displays positioned in this way are referred to as head-down display (HDD). In order to read this display, while driving, drivers must take their eyes off the road ahead; this is unavoidable and would seem to affect driving safety. Zwhalen et al. (1988) pointed out that if a driver's gaze leaves the road for longer than 2 s, then traffic accident risk is significantly increased. This 'attention-away-from-the-road' situation is one of the main factors causing danger on the roads (French, 1990; Wierwille, 1995). CVOSs are still in their infancy in Taiwan. It is important, however, that the logistics industry develops a new visual display interface to convey critical information from these systems to the drivers, before we are faced with the consequences of a major truck accident.

In comparison to the HDD interface, commonly used in the auto industry, the head-up display (HUD) reduces the number and duration of the driver's sight deviations from the road, by projecting the required information directly into the driver's line of vision. This allows drivers to receive information without lowering their gaze, thus avoiding attention gaps that result from them taking their eyes off the road to look down at the information on a HDD (Dingus et al., 1989; Green, 1999). In this way, the driver can easily keep his driving under control (Kiefer, 1991;

Kaptein, 1994), and can quickly respond to information relating to the road environment from the in-vehicle communication system (Iino et al., 1988; Okabayashi et al., 1989).

Considering the reduction in both the number of times drivers take their eyes off the road, i.e. transition from the road to the related visual display interface, (Haines et al., 1980), and re-accommodation time (Larry and Elworth, 1972; Okabayashi et al., 1989; Weintraub and Ensing, 1992), the HUD seems to be a feasible substitution, worth considering, when developing an auxiliary visual display interface for a CVO system.

In comparison to the HDD, most HUD research has focused on providing information on speed limit restrictions, the driver's reactions to accidents and psychological condition. It has also been found that automobile speed is maintained at a more consistent level (± 5 mph within speed limits: Sojourner and Antin, 1990), that drivers were more aware of the speed of their vehicles (Briziarelli and Allan, 1989) and more closely adhered to the posted speed limit, while using an HUD (Rutley, 1975; Kurokawa and Wierwille, 1991). In addition, the speed of drivers having an HUD is, on average, faster than that of drivers having to look head-down at the dashboard (Iino et al., 1988; Kato et al., 1992). However, in the studies of Hooey and Gore (1998) and Kiefer (1991), no significant difference was found in average vehicle speed between the use of head-up and head-down displays.

In terms of responding to unexpected occurrences, the detection rates and response times were comparatively higher and faster when subjects used HUD (Weihrach et al., 1989; Sojourner and Antin, 1990); however, in notable aviation safety studies, reaction times were faster using the HUD only in a low-workload situation. When the load condition was high, on the other hand, the HUD users had longer reaction times when compared to the HDD (Fischer et al., 1980; Iarish and Wickens, 1991; Wickens et al., 1993).

The introduction of new technological displays (either LCD, HDD or HUD) into CVO systems is almost inevitable; the "novelty effect" found in Kiefer's (1991) study (the time a driver spends scanning the speedometer under low-workload conditions was higher during the first session, while successive sessions showed no difference for head-up or head-down) is possible and may have had some effect on the performance of drivers the first time they used the HUD or the HDD in this research. Even though, in Kiefer's study, this novelty effect negatively affected the subject's performance when using the HUD, the questionnaire results showed that 88% of the subjects preferred the HUD to the HDD; 75% of subjects preferred the HUD because they could pay attention to both the road ahead and vehicle speed (Kiefer, 1991), while 14 out of 20 felt the HUD was easier to use (Briziarelli and Allan, 1989). However, an opposite finding indicated that drivers felt that the HUD required a higher mental effort and produced higher mental demand workload ratings (Ward and Parkes, 1994).

Although the HUD has been used widely in the aircraft industry and in the military, (the first operational HUD was installed in the Hawker-Siddeley Buccaneer in 1960), they have only been considered for automobile use since around 1985

(Enderby and Wood, 1992). While many aviation research results could be used as references for this research, considerable differences would have to be taken into consideration, since road traffic conditions are much more complicated and congested than those of the sky; caution is, therefore, warranted when considering these other studies (Kiefer, 1991; Wierwille, 1993).

1.1. Objectives

The people behind one of the Government's ambitious projects, the Taiwan's Future ITS Plans, are becoming aware of CVO systems and understanding that how a vehicle display affects a driver's behaviour is a major hurdle to overcome (i.e. if the system is not user-friendly, it can jeopardize safety).

Ironically, although research into display effect comparisons is one of the main factors in the development of CVO/ITS systems in most other countries, there has been no research on this subject, to date, in Taiwan. This motivated the authors to investigate this pioneering research issue.

This research, using the logistic delivery lorry in-vehicle information system display design, completed in earlier research, will carry out performance evaluation of two types of displays: HDD and HUD. The following research questions will be explored:

1. Given roads with the same driving load, does use of the two displays cause different effects on the performance and workload ratings of drivers?
2. Given roads with different driving loads, does use of the same display cause different effects on the performance and workload ratings of drivers?
3. Given roads with the same driving load, does the use of the same display cause different effects on the performance and workload ratings of drivers because of the arrangement of the display sequence used?

2. Methods

Based on previous research (Liu and Wen, 2004), this study conducted a driving simulation to evaluate the effects of head-up vs. head-down displays. The earlier research used a two-stage procedure in the design of a visual display for CVO drivers on the road. A brief description of this follows.

2.1. Study to design logistics delivery lorry in-vehicle information system display

In the first stage, the authors generated relevant information items/functions for the CVO system via a focused group method: 6 logistic delivery lorry drivers participated in these focused, brain-storming group discussions. Next, a 5-point scale questionnaire survey, with answers going from a scale of 1 (not at all important) to 5 (extremely important), asked 90 haulage drivers to rate the importance of the information items proposed. Factor/correlation analyses of the drivers' opinions

were carried out, grouped as feasible information categories (functions) and the information items within each category listed in order of priority. Then, these results were combined with the human factor guidelines generated by [Green et al. \(1995\)](#), such as General Guidelines for Visual Displays and Navigation Guidelines for Visual Displays (see [Liu and Wen, 2004](#), for details), to design the display interface layout ([Fig. 1](#)) and the necessary CVO information item format for this study ([Fig. 2](#)).

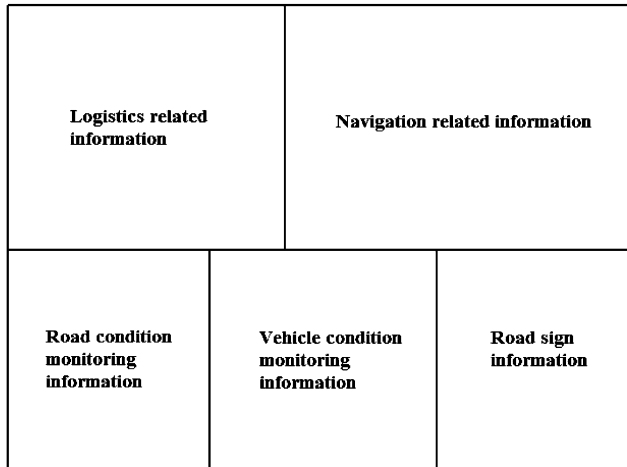


Fig. 1. Screen layout sketch for both head-up and head-down displays.

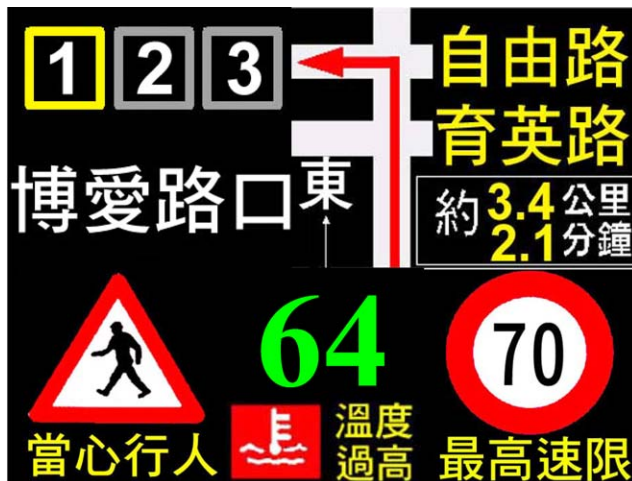


Fig. 2. Example of display with full information content (description of information content can be found in [Table 1](#)).

2.2. Participants

Due to the tight work schedules of the drivers working for the company involved in this research, we asked Taiwan's biggest logistics delivery company to provide 12 drivers to take part in the experiment on a voluntary basis. The group consisted of 11 men and 1 woman, with an average age of 38, an average of 5 years driving experience and having the same qualification levels as determined by the company's annual driving skills/physiological/psychological evaluations. The participant male to female ratio in this study is close to the actual driver population ratio for commercial lorry drivers in Taiwan. The number of participants was based on empirical evidence summarized by Nielsen (1994), which suggested that information elicited from five professional evaluators or users was sufficient to identify around 75% of total usability problems. Participants had to meet the normal requirements for vision of at least 1.0 (or 1.0 after correction); hearing (able to communicate with experimenters); color blindness (able to pass the Ishihara color card blindness test); and they also had to have no experience in the use of a driving simulator or a HUD. After providing informed consent, they were divided into two groups of 6 drivers each. Participants completing the experiment were each paid US\$20.

2.3. Apparatus

2.3.1. STI driving simulator

A STISIM Model 300[®] low cost, fixed base simulator (developed by System Technology, Inc.) was placed in a Volvo DL340 cab (Fig. 3). The controls, e.g. the accelerator, the brake and steering wheel were in exactly the same positions



Fig. 3. Vehicle cab.

as in a real car. The road scenery was projected, using a Plus PJ-020U projector, onto a 120-inch screen, 4 meters in front of the driver. The STI simulator is widely used in research (details can be found at http://www.systemstech.com/SIM_RWA.htm) and the validity of its results is high (Stein, 1990; Allen and Jeffrey, 1995).

2.3.2. Displays

Head-up display. Considering the budget limitations, we used basic principles according to the image projection reported by Tinker et al. (1996), Green (2001) and Horrey and Wickens (2002) to build a HUD that had the ability to communicate with the STI simulator used in this research (as shown in Fig. 4). The information display situation is shown in Fig. 5.

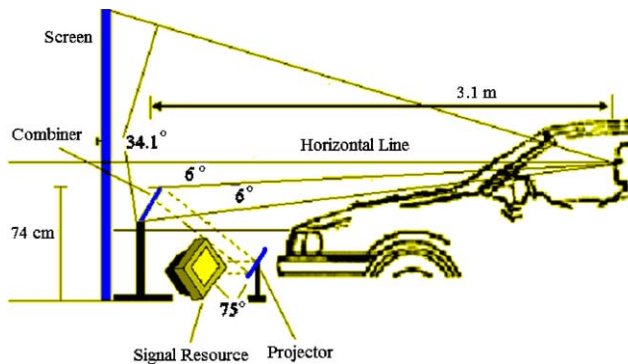


Fig. 4. Layout sketch depicting simulator arrangement, driver, visual scenery screen and HDD.



Fig. 5. Scenery example of HUD from driver's point of view.



Fig. 6. Head-down display.

Head-down display. The aforementioned information, required by the driver when driving, was displayed on an interior 12-inch LCD display, which served as the HDD. This was located near the AC and stereo control panel (Fig. 6). Both the HUD and the HDD contained the same information content, as described in Table 1.

2.4. Tasks

In order to simulate the use of these two different display interfaces, and to record the commercial lorry driver's behaviour and mental stress while receiving the information from the CVO system, tasks were designed around the driver's actions while he was actually on the road, at work. Participants were requested to perform four tasks, concurrently. As correctly and as quickly as possible, and without taking any unnecessary safety risks, they were asked to do the following:

Speed detection/speed maintenance. While obeying the information presented by the system, drivers were asked to keep their speed as close to either 64 km/h (40 mph) or 112 km/h (70 mph) as possible. They were also required to respond to the speed limit signs appearing on the display as quickly as possible (by pushing/releasing the gas pedal and the force exceeding 10% of the original).

Navigation. Participants were asked to follow the system's route guidance information (Table 1) and correctly change roads. As the simulator could not simulate the actual turning, participants were instructed, when turning was required, to say the name of the road and direction in which they wanted to turn, and to turn on the left- or right-turning signal.

Emergency reaction. The display periodically (about every 3 min) issued a road danger warning (e.g. road construction, watch out for pedestrians, etc.) and vehicle monitoring information (e.g. engine temperature too hot); this information was displayed for approximately three seconds before disappearing.

Table 1
Information content for both head-up and head-down displays

Types of information	Examples of information shown in the displays	Descriptions
Logistics		3 delivery points, i.e. destinations 1, 2 and 3 and their corresponding locations
Navigation		Compass direction Turning direction Name of street to turn onto Street name before the turn Distance, e.g. 3.4 km and Time before turn, e.g. 2.1 min
Road conditions		Pedestrian crossing; road construction
Road signs		Speed limits, e.g. 40 and 70 km/h Road curve ahead, e.g. road curving to the left and road curving to the right
Vehicle conditions		Engine temperature too high Speedometer

The driver was required to react by applying the brakes as soon as they detected the warning.

Commercial delivery. The participants carried out delivery work, exactly as they would on a normal working day, with cargo-related information being provided by the in-vehicle information system. This information included

delivery location and delivery receipt number (see Table 1 for details). After arriving at the delivery point, the drivers were asked to tell the experimenter that they had arrived.

2.5. *Driving scenario descriptions*

The driving environment was adapted using the factors considered by Liu (2001), as shown in Table 2, with driving load conditions divided into two levels: high and low.

Each of the two driving load environments was combined with either the head-up or head-down display, creating two scenarios for each load environment. Also, in order to familiarize each participant with the simulator’s displays and have equivalent skills in reacting to the simulator’s instructions, a 5-min training session was developed. Each participant was required to feel comfortable and in control of the simulated vehicle, be able to understand the display information and perform the tasks with no errors, before the actual experiment began.

The participants had to complete one of the display combinations mentioned above with each group driving in one driving load environment; each combination lasted for 20 min. During the test, the driver was required to complete three set-point cargo deliveries (of 5 min each, approx. 15 min total). Following this, they were allowed to drive for about five minutes in a no-cargo delivery work driving environment (the display providing no goods or navigational information—a “pure driving” environment). In the case of emergency events and changes in road speed limits, the scenario followed situations which occur on real roads with the relevant information; this appeared on either a head-up or head-down display (Table 1).

Table 2
Low/high driving load factors

Load factors	Driving load levels	
	Low	High
Lane width (m)	4.1	3.6
Road type	Straight two way lane	Curved two-way lane
Number of easy curves	5 (3100 m radius)	5 (3100 m radius)
Number of sharp curves	0	5 (1500 m radius)
Speed limits	40 mph (64 km/h); 70 mph (112 km/h)	40 mph (64 km/h); 70 mph (112 km/h)
Density of oncoming traffic	Low: average of one vehicle per 550 m	High: average of five vehicles per 100 m
Number of intersections	Average of 28	Average of 80
Density of roadside buildings	Low: 2 buildings for every 2 min driving	High: 20 buildings for every minute driving
Location of roadside buildings	20 m from the roadside	3 m from the roadside

2.6. *Experimental designs*

The research consisted of a 2 (high/low driving load, between-subjects) $\times$ 2 (head-up/head-down display, within-subject) $\times$ 2 (different arrangements of display sequences, between-subjects) mixed-factorial experiment. The 12 participating delivery drivers were divided into two driving load groups (six in a high-load environment and six in a low-load environment); within each driving load group, the six drivers were further divided into another 2 subgroups of 3 drivers each, according to the two different arrangements of display sequences. Each load environment included two scenarios, each having 20 min of driving time: the first consisted of the HUD followed by the HDD; the second consisted of the HDD followed by the HUD.

To avoid a learning effect or other order-related factors which could affect experimental results, the order of displays used was arranged using a counter-balancing method for the participant groups.

2.7. *Data collection*

The drivers' performance data, collected for the comparative study of head-up vs. head-down displays, included both objective and subjective data.

For the objective measurements, we referred to the study conducted by Liu (2001), which measured driver behaviour. This included average speed/variance of speed (ft/s); lateral lane position variation (ft); longitudinal acceleration variation (ft/s²); lateral acceleration variation (ft/s²); throttle input variation (ft/s²); steering wheel angle variation (deg); and brake pedal input (ft/s²). The reaction time (s) of releasing/pushing the throttle/brake pedals was also collected for the detection of reaction to the speed limit changes. For response to urgent events, the reaction time (s), and the reaction accuracy rate (%) in applying the brake pedal were collected. The data related to driving behaviour, speed limit maintenance/detection and emergency response time were collected via the simulator. In addition, accuracy in navigational turning correctness (%) and goods delivery accuracy rate (%) were collected by means of a pre-prepared checklist, recorded by the experimenter.

For subjective measurements, a 5-point scale measurement questionnaire was used. After the experiment, using both types of display, was completed, participants were asked to complete the questionnaire to assess the mental pressure they had felt during the simulation; this included time stress—insufficient time to finish reading the display information (or the roadside information); mental stress—feeling tired or frustrated when receiving information on the display; visual stress—visual overload, caused by excess information on the display (or the roadside information); degree of interference—difficulty in perceiving road side information because of display interference. A rating of 1 on the 5-point scale indicated a low stress load; 3 indicated a medium stress load; and 5 indicated very high stress load.

All the data collected were processed and reduced. Statistical analyses, using ANOVAs and Kruskal–Wallis, were conducted with SPSS v.10.0[®] to understand the

differences in the CVO system, while using either HUD or HDD, with different display sequence combinations and in different driving environments. The significance level α was set to 0.05.

3. Results

For commercial goods delivery and navigational tasks, the results showed no significant difference between drivers using HUD and HDD, regardless of display arrangement sequence and driving load conditions. Overall, the drivers, using both displays, completed the two tasks with very high rates of accuracy (goods delivery task: 100% for both displays; navigational task: head-up: 95.8%; head-down: 96.3%).

We present the results of our analysis below, based on the three objectives this study set out to achieve.

3.1. Comparison of HUD and HDD performance under the same driving load

Table 3 indicates the performance measures for drivers using HUD vs. HDD in each driving load condition. As shown, in a high-driving load environment, there was a significant difference in performance, for different displays, in terms of driver behaviour and reaction time. The first item, delivery of commercial goods, the HDD was significantly higher in terms of “speed variation” than the HUD.

This result illustrated that, especially for first time use in high-load road conditions, it was more difficult to maintain speed when using an HDD than when using an HUD. The reaction time for detecting speed limit changes from 40 to 70 mph and from 70 to 40 mph, and the reaction time for urgent events, resulted in statistically significant differences between HUD and HDD usage: HUD vs. HDD: 1.2233 vs. 1.4617; 0.99 vs. 1.2089; 1.3235 vs. 2.3274; respectively. It was clear that when participants used HUD, their reaction times were shorter than when using HDD.

Table 3
Performance measures for HUD vs. HDD in driving load conditions

Load	Performance measures	HUD	HDD	<i>F</i> (1, 10)	<i>p</i> -Value
High	Speed variation (ft/s)	10.153	20.396	6.899	0.025
	RT for speed limit sign changes (s)				
	From 40 to 70 mph	1.2233	1.4617	5.000	0.049
	From 70 to 40 mph	0.99	1.2089	5.826	0.036
	RT for emergency response (s)	1.3235	2.3274	9.482	0.012
Low	RT for emergency response (s)	1.0073	1.8684	22.269	0.01

In terms of stress load, participants were asked: “Was the information appearing on the roadside easy to see?” (Score 1: very difficult to 5: very easy). There was a slight difference between the HUD and HDD results [$\chi^2(1) = 3.667$, $p = 0.056$]. Although this difference was not statistically significant, the roadside information using the HUD (2.000) was slightly more difficult to make out than when using the HDD (1.500).

In a low-load environment, a significant difference was found between the use of HUD and HDD in terms of reaction time to urgent information. The average reaction time of 1.0073 s for the HUD was noticeably lower than for the HDD (1.8684).

3.2. Comparison of the same display under different driving loads

Table 4 reveals the results comparing drivers' performances, using one of the two displays, between the two driving load conditions. As can be seen, using HUD, in a high/low driving load environment, the average lateral lane position (the smaller the number, the closer to the centre line), lateral acceleration variation (the larger the variation, the higher the demand for the driver's attention), and steering wheel angle variation (the larger the variation, the poorer the vehicle control) showed significant differences; performance on low-load roads was better than on high-load stretches of road.

In addition, while using an HUD in a low-load driving environment, drivers were able to respond more quickly to warning information than on high-load stretches (1.0073 s for low-load compared to 1.3235 s for high load).

Similar results were obtained using an HDD. Lateral acceleration variations and steering wheel angle variations showed better performance on low-load road than on high-load road conditions.

In low-load driving conditions, reaction time to the 40 mph speed limit sign was 1.1425 s using an HDD. On the other hand, in high-load driving conditions, more

Table 4
Performance measures for each display in different driving load conditions

Display	Performance measures	Load		$F(1, 10)$	p -Value
		High	Low		
HUD	Mean lateral lane position (ft)	5.8474	7.0946	9.511	0.012
	Lateral acceleration variation (ft/s ²)	4.3234	1.2142	9.696	0.011
	Steering wheel angle variation (deg)	1.9166	0.6305	8.622	0.015
	RT for emergency response (s)	1.3235	1.0073	10.353	0.0001
HDD	Lateral acceleration variation (ft/s ²)	5.5803	1.3112	6.891	0.025
	Steering wheel angle variation (deg)	2.8828	0.7221	11.170	0.007
	RT for emergency response (s)	2.3274	1.8684	5.582	0.040
	RT for speed limit sign changes (s) From 40 to 70 mph	1.4617	1.1425	6.034	0.030

time—1.4617 s—was required. There was also a significant difference in reaction time for an emergency event. Reaction time in high-load conditions was higher (2.3274 s), while in low-load conditions reaction time was only 1.8684 s.

3.3. Comparison of arrangements of display sequence in different driving environments

Table 5 shows comparisons, under different driving load conditions, for the two arrangements of display sequence.

On a high-load road, when drivers used the HUD first, a higher variation in speed maintenance and a slower reaction to urgent events occurred than for those using the HUD later. The visual stress, in receiving roadside information, felt by the driver who used the HUD first, was higher than that felt by the driver who used HUD later (3.0 vs. 2.0); 33.33% decrement in stress ratings. On the other hand, drivers who used the HDD first rated their visual stress, in receiving the HDD information, higher than those using the HDD later (3.667 vs. 2.333); the decrement was 36.38%.

In low-load driving environments, although neither of the display sequences caused significant differences in the drivers' driving behaviours, they did affect the drivers' perception of psychological workload. Using the HUD in the first section had a more significant impact, in terms of frustration in receiving roadside information, than using the HUD in the later section (3.0 vs. 1.333); the frustration load decrement reached 55.57%.

In low-load environments, the use of HDD first caused markedly more visual stress, time stress and workload frustration ratings than when using it last (2.667 vs. 1.333, down by 50.02%; 3.333 vs. 1.667, down by 49.98%; 3.000 vs. 1.333, down by 55.57%, respectively).

4. Discussion

The results, showing no significant difference for commercial goods delivery and navigational tasks between the two displays, indicate that if the information is displayed for a long enough time (e.g. for goods delivery and navigational information in this study), accurate information assimilation can be achieved using either the head-up or head-down displays. This result supports the findings of Hooley and Gore (1998), which state that no clear difference exists between the navigation accuracy rates of HUD and HDD. Using both displays, the driving behaviours of the drivers (i.e. variances in lateral acceleration and steering wheel angle) and their reaction times deteriorated as the driving load increased (Table 4).

Driver response to emergency-related information (i.e. watching out for pedestrians, road construction, speed limits and engine temperature too high) showed a clear difference in reaction time between high- and low-level road situations for both types of displays. Specifically, the HUD helped drivers react more quickly to the in-vehicle CVO system's warning and road prompting information than the HDD did (Table 3). This result confirms the findings of Okabayashi et al. (1989) and Wickens et al. (1993).

Table 5
Performance measures for display sequences under different driving load conditions

Driving loads	Performance measures	First on the HUD	Later on the HUD	<i>F</i> (1, 5)	<i>P</i> -Value	χ^2 (1)	<i>p</i> -Value
High	Speed variance (f/s)	12.3494	7.9572	15.327	0.017	5.0	0.025
	RT for urgent response (s)	1.4369	1.21	21.985	0.009		
	Visual stress in receiving roadside info.	3.0	2.0				
		First on the HDD	Later on the HDD				
	Visual stress in receiving display info.	3.667	2.333			4.433	0.033
Low		First on the HUD	Later on the HUD			4.50	0.034
	Frustration in receiving roadside info.	3.0	1.333				
		First on the HDD	Later on the HDD				
	Visual stress in receiving roadside info.	2.667	1.333			4.121	0.047
	Time stress in receiving display info.	3.333	1.667			4.091	0.043
	Frustration in receiving display info.	3.0	1.333			4.50	0.034

In terms of driving behaviour, the “speed variations” and “response to speed limit signs” that occur when using the HUD in a high-load environment are lower and faster, respectively, than when using the HDD; this shows that when the driving load is high, the driver’s effectiveness in receiving information is still higher from the HUD than from the HDD (Table 4). A similar conclusion was reported by Iino et al. (1988) showing that, in all experimental environments, the HUD reading time is shorter than the HDD. By reducing the amount of time the driver’s eyes are on the display, and therefore off the road, vehicle control improved and vehicle drift is reduced. It can be concluded, therefore, that an in-vehicle information system using an HUD as the visual display, allows the driver to quickly assimilate information and, at the same time, pay attention to related roadside information; as a result, driver stability improves.

However, with respect to “average speed”, this research found no major differences between the use of the HUD and HDD on speed maintenance. This result is the same as that of Hooy and Gore (1998); is different from that reported by Kurokawa and Wierwille (1991), which showed a clear effect on speed maintenance performance, using different displays, and that of Rutley (1975), which showed that HUD allows for easy speed maintenance. The differences from these two studies can perhaps be attributed to the fact that the participants in our study were professional delivery lorry drivers, whereas the previous research used ordinary drivers. With the drivers used in this study, driving was their profession, and so they were able to maintain car control using both the displays; the results clearly show that there was no major effect on speed maintenance when using these displays.

There was no significant difference, overall, in the drivers’ psychological stress load, when comparing the HUD and the HDD, although the HUD scored a slightly “more difficult” rating for the received of roadside information than the HDD, when the driving load was high ($p = 0.056$). However, when comparing the first use and later use of the display combinations, the results showed that, in general, regardless of the display used, the “later” use in the subjective workload ratings (i.e. visual stress, time stress, mental frustration) were all clearly less than for “first” use (Table 5). The “novelty effect” of using new high-tech products (i.e. HUD and HDD) proposed by Kiefer (1991) is, to some extent, proved in this study.

To briefly summarize the results presented in Section 3.3: in most situations, the improvements from first use to later use of the HDD subjective ratings were greater than the HUD results, showing that HDD in both high and low-load road conditions improved more with the later use (Table 5). These findings seem to indicate that if in-vehicle information systems want to introduce HDD, training materials would be required to familiarize the drivers with the system. In addition, drivers using the HUD first consistently felt pressure when receiving roadside information, while drivers who used the HDD first, especially in the low-load road conditions, consistently felt stress when receiving information on the display. The authors believe that these consistent subjective workload ratings indicate that the HUD, to some degree, affects or distracts the driver in receiving road information, whether the driving load is high or low. With HDD use, it is necessary for the driver to switch his/her view from the roadway to the display, and

vice versa, thus increasing his/her psychological stress in perceiving display information.

Results of significant stress ratings for first time use of the HDD, between high- and low-driving loads (1 for high load vs. 3 for low load), indicate that drivers perhaps decreased (or ignored) their display viewing frequency when their driving loads increased, thus reducing their stress.

Finally, with regard to the number of the participants: the current study consisted of 12 subjects, 6 for each driving load condition. Gawron (2000, pp. 5–6) postulated a formula for calculating the “effect size” [effect size = (absolute value of the performance difference between using the HUD and the HDD)/standard deviation of that performance measure]. According to this formula and data for emergency response reaction time from Table 4, we obtained the effect size on hard driving load conditions: $(2.3274 - 1.32345) / 0.518 = 1.9381$. Then “based on over 100 years of experimentation and statistics” (Gawron, 2000, p. 6, first paragraph), a curve showing the number of subjects needed as a function of the effect size can be read as the number of subjects (~5) required for each condition. This study fitted the suggested number of subjects.

5. Conclusions

Generally, as driving workload increased, driver’s performance, using both displays, was negatively affected; in the high-driving load conditions, the HUD produced better speed control and faster reaction to both speed limit signs and urgent events than the HDD. Consequently, the authors believe that a driver using an HUD is more cautious and aware of the road environment and therefore, more obedient to traffic regulatory signs. Because of this, CVO systems designers should consider using an HUD to display emergency and traffic regulation information.

As well as considering objective performance, a driver’s subjective feelings towards the display can serve as another major factor in making this product commercially acceptable. The “novelty effect” finding should make the CVO systems designers pay attention to the training material(s) developments.

Owing to the interference problem associated with the HUD, however, researchers should further investigate the issue of information clutter. Future studies may also consider adding auditory signals to the display modality to improve the reception of CVO-related information, especially in high-workload situations.

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